HAMDARD UNIVERSITY BANGLADESH

FACULTY OF BUSINESS ADMINISTRATION

MGT 417: Industrial and Operational Management

Semester: Spring 2024

12/06/2024 **Midterm Examination** Time: 1.5 Hours

Total Marks: 10 x 3=30

Instructions:

- All parts of each question must be answered sequentially.
- The figures on the right indicate respective marks.
- Answer any three (3) of the following questions.

01.	a.	What is an Industry? What are the elements of Industry Environment?	**05**
	b.	Discuss Product Life Cycle in detail with relevant model.	**05**
02.	a.	"The Three Levels of Management Decision Making has direct impact on Industrial Operations Management." Determine whether you agree with the statement.	**05**
	b.	Distinguish between Product and Service with relevant example.	**05**
03.	a.	"Plant Location along with Plant Layout result in a faster supply chain."- Defend your position on the matter.	**05**
	b.	Why more focus is put on the Feedback part of the Simple Block Diagram?	**05**
04.	a.	What is Supply Chain Management?	**02**
	b.	What are the functions of Operations Management in Supply Chain Management?	**08**

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE325: System Analysis and Design

Batch: 21st

Semester: Spring 2024

Date: 01/07/2024

Mid-Term Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions of the followings. $5 \times 6 = 30$

1. (a) Explain the system designers' view of information system. 3
   (b) Describe the system builders' view of Knowledge of system design. 3

2. (a) Write the guidelines for developing Use Case Diagrams. 3
   (b) List some business questions that can help a system analyst to identify the business objectives of a system. 3

3. (a) Why feasibility analysis is required to initiate a project? 2
   (b) Identify the observable elements of STROBE. 4

4. (a) Write the advantages and drawbacks of Bring Your Own Device (BYOD) option in system development. 3
   (b) List the situation when each of the following analysis can be applied: 3
      i. break-even analysis
      ii. cash-flow analysis
      iii. present value analysis

5. (a) Describe the steps for planning the interview. 3
   (b) Illustrate the pyramid structure for interviewing. 3

6. Zihad owns a frozen-foods company and wants to develop an information system for tracking shipments to warehouses. The information of the development process is given below: 3

Task	Must Follow	Time (Weeks)	Crash Time	Cost Per Week
A	None	5	3	$500
B	None	3	2	$300
C	B	2	2	$400
D	A, C	3	2	$750
E	C	4	3	$200
F	D	4	2	$250
G	E, F	5	3	$400
H	G	3	2	$380

(a) Draw the PERT diagram for the system and identify the critical path. 3
(b) If Zihad expedites the project as much as he can, calculate the saving amount and the minimum time that the project will take. 3

7. (a) What is meant by scaling in questionnaires? Explain different forms of measurement scales used to interview. 3

(b) What is meant by sampling in information gathering? What are the reasons for system analyst would want to sample data or select representative people to interview? 3

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HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE321: Theory of Computing

Batch: 21st

Semester: Spring 2024

Date: 26/06/2024 **Mid-Term Examination** Time: 1 hour 30 minutes

Total Marks: 30

Answer any **five** questions. **Marks: 5 × 6 = 30**

1. (a) Write the uses of automata. 2
   (b) State the important issues that define automata. 4

2. (a) Draw an NFA for the string **a*** + **(ab)***. 2
   (b) By using inductive proof technique, prove that 4

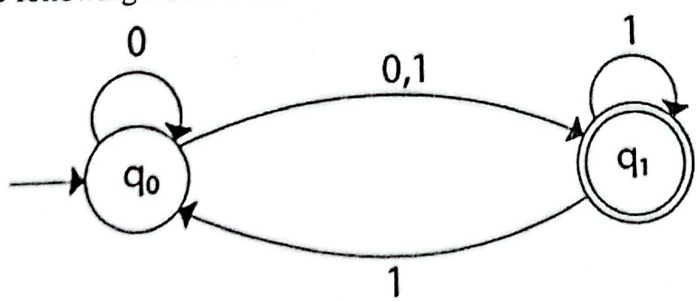
$$1 + a + a^2 + \cdots + a^n = \frac{1 - a^{n+1}}{1 - a}$$

3. (a) Compare between Σ^+ and Σ^*, when Σ={abc}. 2
   (b) Show that for DFA, δ: $Q\times\Sigma \rightarrow Q$ and for NFA, δ: $Q\times\Sigma \rightarrow 2^Q$. 4

4. (a) With proper diagram, differentiate between NFA and DFA. 3
   (b) Write short notes on 1×3=3
      i) Alphabet
      ii) Power of Alphabet
      iii) Concatenation of string

5. Design DFA for the followings 2×3=6
   a) ends with '**abc**'
   b) Substring of '**abc**'

6. Convert the following NFA to a DFA: 6



7. Design a DFA accepting the set of all strings ending with **00** over the alphabet {0,1}. Show that **1110011001** is not accepted by the system. 6

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT323
Mid-Term Examination
Date: 6/29/2024

Total Time: 1Hours:30 Minutes.

Course Title: Numerical Analysis
Semester: Spring 2024
Total Marks: 30

Answer any three (03) of the following questions: $3 \times 10 = 30$

1. (a) Explain the operator E, ∇ and Δ [3]
   (b) Find the approximate root of the equation $f(x) = x^3 + x^2 - 1 = 0$ by using bisection method$[a = 0, b = 1]$. [3.5]
   (c) Prove the given relation [3.5]
   (i) $(1 - \nabla)(1 + \Delta) = 1$.
   (ii) $E = 1 + \Delta$.
2. (a) Write down a short note of false position method. [2]
   (b) Find the approximate root of the equation $f(x) = x^3 - 3x - 5 = 0$ by using false-position method. [4]
   (c) Find the approximate root of the equation $f(x) = 2x^3 - 3x - 6 = 0$ by using Newton-Raphson method. [4]
3. (a) State the Lagrange's interpolation formula for $n + 1$ data. [1]
   (b) Find the value of the function $y = f(x)$ for $x = \alpha = 1.5$ and x for $f(x) = -10$ from the following table [using Lagrange's interpolation formula]: [9]

x	0	1	3	4
$y = f(x)$	-12	0	12	24

4. (a) Find the number of student who obtained less than 42 marks form the following table: [4]

Marks (x)	0–19	20–39	40–59	60–79	80–99
Population $f(x)$	41	62	65	50	27

   (b) What is interpolation? Find the value of the function $y = f(x)$ for $x = 12$ from the following table: [6]

x	5	11	27	34	42
$f(x)$	23	899	17315	35606	68510

5. (a) Write the procedure of Gauss seidel method to solve system of equations [3]
   (b) Solve the following system of equations by using Jacobi and Gauss seidel method: [7]

$$10x + y - 2z = 7.74$$
$$x + 12y + 3z = 39.66$$
$$3x + 4y + 15z = 54.8$$

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 323 : Operating System

Batch: 21st | Semester: Spring 2024
Time: 1 Hour 30 Minutes
Date: 10/06/2024 | **Midterm Examination** | Total Marks: 30

Answer any five questions. $5 \times 6 = 30$

1.	(a)	Define Operating System. Illustrate where the Operating System fits in a Computer System.	3
	(b)	What are the roles of an Operating System? Briefly Explain.	3
2.	(a)	Mention different types of kernel-based architectures available for Operating Systems.	2
	(b)	Explain the Interrupt-driven I/O cycle in Computer System Organization.	4
3.	(a)	State the Dual-Mode operation of CPU?	2
	(b)	Explain the transition from user to kernel mode with necessary figure.	4
4.	(a)	Explain the functionality of the Fork system call.	2
	(b)	Write the steps in making the following System Call. count = read(fd, buffer, nbytes);	4
5.	(a)	Explain the lifecycle of a Process.	3
	(b)	Explain Context Switching with necessary figure.	3
6.	(a)	Mention some principal events that cause processes to be created.	2
	(b)	Explain the thread usage using a Web Server.	4
7.	(a)	Why does each thread have it's own stack?	3
	(b)	Briefly describe the Hybrid implementation of Thread.	3

FACULTY OF BUSINESS ADMINISTRATION

MGT 417: Industrial and Operational Management

Semester: Spring 2024

02/10/2024

Final Examination

Time: 2.00 Hours

Total Marks: 10 x 4=40

Instructions:

- All parts of each question must be answered sequentially.
- The figures on the right indicate respective marks.
- Answer any four (4) of the following questions.

01. a. What is Project Management and what are its key elements? **05**

b. Suppose you are to build a software from Scratch as a Project Manager. What are the Project Lifecycle Phases that you will have to go through as part of this Project. Also mention the challenges that you may face in managing the project. **05**

02. a. What do you understand by Quality Circle. Discuss its impact on a Manufacturing based business. **05**

b. Contrast between Product and Project Management with Relevant Diagram. **05**

03. a. **05**

Activity	Time in Weeks
1-2	8
1-3	5
2-4	9
3-5	7
2-5	5
4-6	3
5-6	6
6-7	10
5-7	9

Find the time to complete the project using Critical Path method.

b. Differentiate between PERT and CPM. **05**

04.

Week	1	2	3	4	5	6	7	8
Sales	39	44	40	45	38	43	39	?

a. Forecast Sales using Simple Moving Average Method and Discuss why it is not trustworthy all the time. 05

b. Forecast Sales using 4-week Weighted Moving Averages with Weights 0.4, 0.3, 0.2, 0.1. 05

05 a. Why is Project Management Principles like Agile or Scrum is used during Software Development? 05

b. Discuss How Just-in-Time work with relevant Diagram. 05

06 Customers arrive randomly at a Spa center with a single service giver. The average rate of customers per hour to the center is 7 per hour. Each customer requires 8 minutes of session each. Having continuous flow of the customers, the only service giver is thinking whether there is any need for another service giver. The arrival process is Poisson, and the rate can be assumed to be negative exponential. Calculate: 10

i. Total Time in the System or Average waiting time in the Spa center

ii. Average Waiting time in line

iii. Average number of Customers in the System

iv. Average Number of Customers in Line/ Queue Length

v. Utilization Rate of the facility or Probability of the System being Busy

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE325: System Analysis and Design

Batch: 21st | Semester: Spring 2024

Time: 2 Hours

Date: 30-09-2024 **Final Examination** Total Marks: 30

Answer any **five** questions of the followings. $5 \times 8 = 40$

1. (a) What are the factors to consider when choosing output technology? 4

   (b) Assume that, the proportion of the population having the attribute $p = 0.05$, acceptable interval estimate $i = \pm 0.02$. Now, calculate the sample size if the confidence level is 90%. 4

Confidence Level	Confidence Coefficient (z value)
99%	2.58
98	2.33
97	2.17
96	2.05
95	1.96
90	1.65
80	1.28
50	0.67

2. (a) List the common errors made when drawing data flow diagrams. 3

   (b) Draw a data flow diagram for payroll system. 5

3. (a) Differentiate between logical and physical data flow diagrams. 4

   (b) Define data dictionary. List the components of it. 4

4. (a) What are goals of producing process specifications? 2

   (b) What are the ways to represent process logic by using structured English? 3

   (c) Suppose a store wanted to illustrate its policy on noncash customer purchases. There are three conditions (sale under \$50, pays by check, and uses credit cards) has only two alternatives. The two alternatives are Y (yes, it is true) or N (no, it is not true). Four actions are possible: 3

      i. Complete the sale after verifying the signature.
      ii. Complete the sale. No signature needed.
      iii. Call the supervisor for approval.
      iv. Communicate electronically with the bank for credit card authorization.

   Rules are the combinations of the condition alternatives that precipitate an action. Four rules are:

      i. IF N (the total sale is NOT under \$50.00)
      ii. IF Y (the customer paid by check and had two forms of ID)
      iii. IF N (the customer did not use a credit card)
      iv. DO X (CALL THE SUPERVISOR FOR APPROVAL)

   Now, generate the decision table for illustrating a store's policy of customer checkout with four sets of rules and four possible actions.

5. (a) Describe the way to reduce the redundancy of data being entered. 2

(b) Define the term bottleneck with example as it applies to data entry. 2

(c) Explain eight general guidelines for proper coding. 4

6. (a) Define data normalization in system design. List the steps of data normalization. 2

(b) Write the differences between data warehouses and traditional databases. 3

(c) Explain the integrity constraints that can apply to a database. 3

7. (a) What are the objectives of creating an effective database for system design? 3

(b) Define Entity Relationship (ER) diagram. Draw an entity-relationship diagram for patient treatment. 5

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE321:Theory of Computing

Batch: 21st

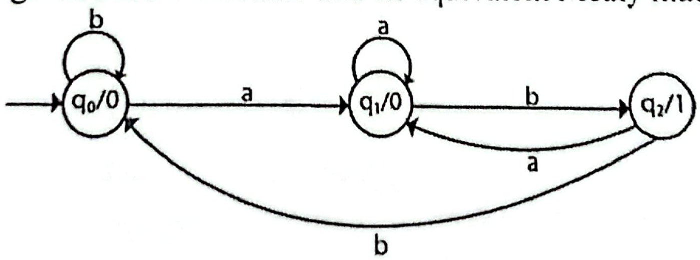
Semester: Spring 2024

Date: 28/09/2024 **Final Examination** Time:2 Hours

Total Marks: 40

Answer any **five** questions. **Marks: 5×8 = 40**

			Marks
1.	(a)	Define Mealy machine with mentioning its possible tuples.	2
	(b)	Design a Mealy machine for a binary input sequence such that if it has a substring 101, the machine outputs A, if the input has substring 110, it outputs B, otherwise it outputs C.	6
2.	(a)	Design a PDA for accepting a language $\{a^n b^n \mid n>=1\}$.	4
	(b)	Design a Moore machine with the input alphabet {0, 1} and output alphabet {Y, N}, which produces Y as output if input sequence contains 1010 as a substring otherwise, it produces N as output.	4
3.	(a)	Define Pushdown Automata (PDA).	2
	(b)	Derive the string "aabbabba" for **leftmost** and **rightmost** derivation using a CFG given by, $S \rightarrow aB \mid bA$ $A \rightarrow a \mid aS \mid bAA$ $B \rightarrow b \mid bS \mid aBB$	3×2=6
4.	(a)	Write a short note on Context Free Grammar (CFG).	2
	(b)	If G is the grammar $S \longrightarrow SbS \mid a$, show that G is ambiguous.	6
5.	a)	Differentiate between Moore and Mealy machine.	3
	b)	Convert the given Moore machine into its equivalent Mealy machine.	5



6. Suppose ,

$S \rightarrow (L) \mid a$

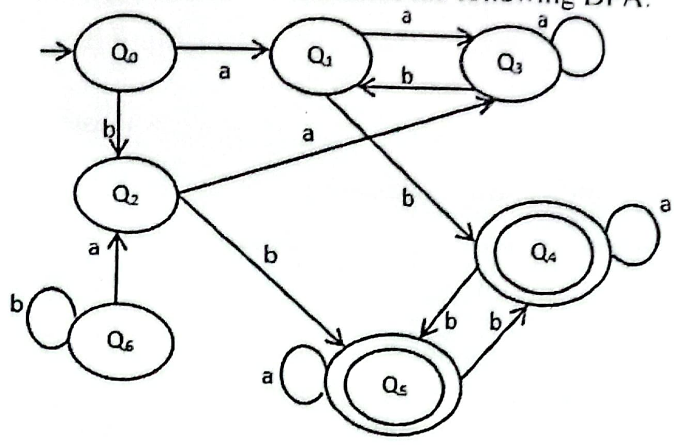
$L \rightarrow L, S \mid b \mid s$

Find **leftmost** derivation for — 4×2=8

(i) (a , (a , a))

(ii) (a , ((a, a) , s))

7. Using the DFA minimization rules, minimize the following DFA: 8



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT323
Final Examination
Date: 10/7/2024

Course Title: Numerical Analysis
Semester: Spring 2024

Total Time: 2 Hours.

Total Marks: 40

Answer any four (04) of the following questions: $4 \times 10 = 40$

1. (a) Write the importance of Numerical Analysis. [3]

   (b) Find the approximate value of the integral $I = \int_{3}^{4.4} \ln(x)dx$ by Trapezoidal rule and Simpson 1/3 rule. Find the exact value $I = 1.823$ and error. [7]

2. Find the approximate value of the integral $I = \int_{1.2}^{1.6} \left(x + \frac{1}{x}\right) dx$ by Simpson 1/3 Weddle's rule and Simpson 3/8 rule. If the exact value is $I = 0.848$, then find the error. [10]

3. (a) Using Euler method solve $\frac{dy}{dx} = 1 + xy$ with $y(0) = 2$. Find the value of $y(0.4)$. [3]

   (b) Using second and third order Runge-Kutta method, find the value of $y(0.4)$ the following first order differential equation $\frac{dy}{dx} = xy + y^2; y(0) = 2$ and compare both methods. [7]

4. (a) Find the approximate value of y for $x = 0.2$ by using fourth order Runge-Kutta method where $\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 + y^2 = 0; y(0) = 1; \frac{dy(0)}{dx} = 0$, taking $h = 0.1$. [10]

5. (a) Write the procedure of Newton-Raphson method to solve nonlinear system of equations. [3]

   (b) Solve the system of non-linear equations by fixed point method. [7]

$$x^2 + y = 4$$

$$y^2 + x = 7$$

6. Find the approximate root of the following system of non-linear problem by using Newton-Raphson method: [10]

$$x^2 - y^2 = 4$$

$$x^2 + y^2 = 8$$

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 323 : Operating System

Batch: 21st | Semester: Spring 2024

Time: 2 Hours

Date: 25/09/2024 | **Final Examination** | Total Marks: 40

Answer any **five** questions. $5 \times 8 = 40$

1. (a) Define Inter Process Communication. Differentiate between Independent and Cooperating Process. 3

   (b) Explain the fundamental models of Inter Process communication with necessary figures. 5

2. (a) Define the followings: 4x1=4
   i. Race Condition
   ii. Critical Region
   iii. Mutual Exclusion
   iv. Busy Waiting

   (b) State the requirements for the solution to Race Conditions in the Operating System. 4

3. (a) What is the Strict Alternation solution to mutual exclusion? What are its limitations? 4

   (b) Define Semaphore. Briefly explain the solution for the Producer Consumer problem using **Semaphores**. 4

4. (a) Differentiate between Preemptive and Non-preemptive scheduling. 2

   (b) Consider the following table for 5 jobs arriving at time zero in the given order: 6

Job	Burst Time
1	15
2	5
3	25
4	10
5	20

Now, find out which algorithm among FCFS, SJF, and Round-Robin with quantum 8 would give the minimum average waiting time.

5. (a) Define Deadlocks in Operating System. State the necessary conditions for deadlocks. 3

(b) Consider the following system with four types of resources (A, B, C, D) and three processes (P1, P2, P3). Apply the deadlock detection algorithm to determine if there is a deadlock in this system. If there is no deadlock, show how each process can complete its execution. 5

$$E = (6 \quad 4 \quad 5 \quad 3) \qquad A = (3 \quad 1 \quad 2 \quad 1)$$

$$C = \begin{pmatrix} 2 & 0 & 1 & 1 \\ 1 & 2 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{pmatrix} \qquad R = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 2 & 0 & 2 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

6. (a) Explain safe and unsafe states in relation to deadlock. 3

(b) State the Banker's algorithm for checking whether a state is safe or unsafe. 5

7. (a) Differentiate between Semaphores and Monitors. Mention the problems with Semaphores and Monitors. 4

(b) What are Barriers in Operating System? Explain how Barriers can synchronize groups of processes with an example. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 327: Computer Peripherals & Interfacing

Batch: 21st

Semester: Spring 2024

Date: 19/07/2024 **Final Examination** Time: 2 hours

Total Marks: 40

Answer any **five** questions from the following: 5x8 = 40

1	(a)	Outline the instructions of a digital computer.	3
	(b)	Illustrate the connection between the interface unit and the highway in a diagram.	5
2	(a)	Define Memory Mapped I/O and discuss its advantages and disadvantages.	4
	(b)	Describe the process of DMA transfer in a computer system and explain the role of the DMA Controller and its connections.	4
3	(a)	Differentiate between serial and parallel interfaces.	4
	(b)	Explain the concept of a Synchronous Serial Interface (SSI).	4
4	(a)	Explain the functions of the bus management lines in GPIB (IEEE 488).	4
	(b)	Provide an overview of the uses for 7-segment and 9-segment displays.	4
5	(a)	Describe the primary components of a hard disk.	4
	(b)	State the differences between FAT32 and NTFS.	4
6	(a)	Provide an explanation of the construction of gas-discharge displays.	4
	(b)	Explain the various components found in a Network Interface Card.	4
7	(a)	Draw the block diagram of the PPI 8255.	3
	(b)	Outline the operating modes of the PPI 8255.	5